2024 Bi H2 Q15

Section: Metabolism and Survival

Topic: Adverse Conditions

Question Summary

Essay question: describe how animals survive adverse conditions. Answers should cover physiological, behavioural, and ecological strategies such as dormancy, hibernation, aestivation, daily torpor, and migration. Definitions, examples, and biological advantages must be included.

Worked Solution

- Animals survive periods of environmental stress (for example, low temperature, drought, or food shortage) by reducing metabolic activity or by moving to more favourable environments.

- **Dormancy** is a period when metabolic rate, heart rate, and body temperature decrease to conserve energy. It can be **predictive**, occurring before conditions deteriorate (for example, hedgehogs preparing for winter), or **consequential**, when animals enter dormancy after conditions become severe (for example, snails in drought).

- **Hibernation** occurs in some mammals such as bats and hedgehogs during winter. Body temperature and heart rate drop markedly; stored fat is used slowly for respiration. The animal remains inactive until temperature and food availability increase.

- **Aestivation** occurs in hot, dry periods to avoid desiccation and overheating. Examples include desert snails and lungfish that burrow into moist mud and form a protective cocoon.

- **Daily torpor** is a short-term reduction in activity and metabolic rate, often seen in small birds such as hummingbirds that enter torpor overnight to conserve energy when food is scarce.

- **Migration** is an active survival strategy where animals move to areas with more favourable conditions. It requires energy expenditure but allows continued feeding and reproduction. For example, swallows migrate from Europe to Africa to avoid winter food shortages.

- These survival strategies allow animals to maintain population stability despite environmental stress, by either conserving energy or relocating to avoid it.

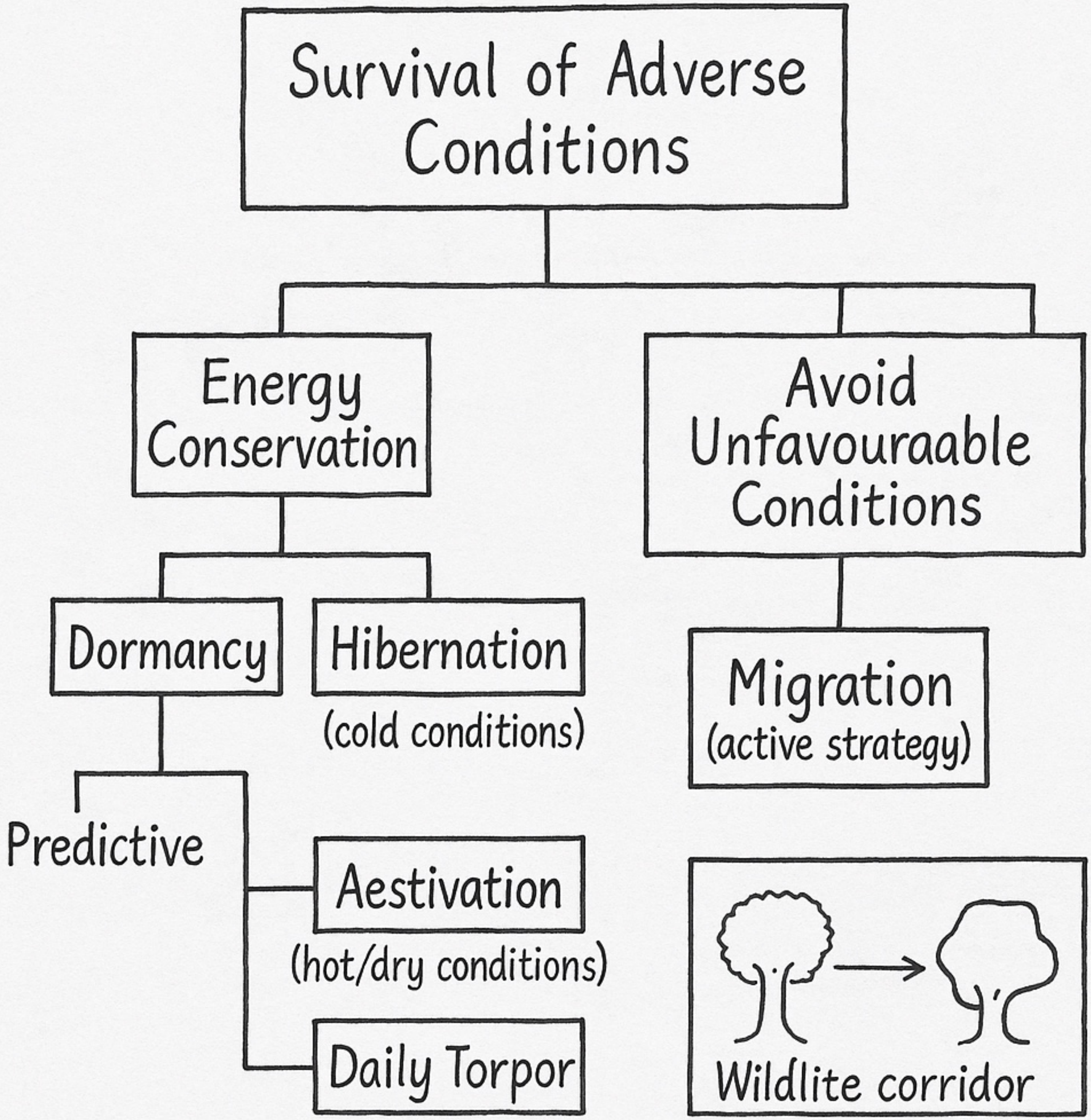
Final Answer

-> Dormancy: metabolic rate decreases; may be predictive or consequential
-> Hibernation: response to cold, e.g. hedgehogs, bats
-> Aestivation: response to heat/drought, e.g. lungfish, snails
-> Daily torpor: short-term energy conservation, e.g. hummingbirds
-> Migration: active movement to favourable conditions

-> All improve survival by conserving energy or avoiding stress

Revision Tips

- Distinguish between predictive and consequential dormancy.
- Include named examples for each strategy.
- Link reduced metabolic rate to energy conservation.
- Migration is energy-intensive but allows feeding and reproduction year-round.


Survival of Adverse Conditions
Energy Conservation
Avoid Unfavouraable Conditions
Dormancy
Hibernation
(cold conditions)
Migration (active strategy)
Predictive
Aestivation
(hot/dry conditions)
Daily Torpor
Wildlite corridor